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in
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Privacy awareness in point clouds using AI-based object detection

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Abstract

Περίληψη

Acknowledgements

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place.

If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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1 Introduction

1.1 Motivation and Problem Statement

1.2 Thesis Structure

Chapter 2

Chapter 3

Chapter 4

Chapter 5

Table 1.1 refers to the results in Figure 4.1.

	Name	Task	Type
1	Tom	Seek	Cat
2	Jerry	Hide	Mouse

Tab. 1.1: This is a test table

2 Background and Related Work

2.1 Background

2.1.1 Background Information and Terminology

Point clouds are a set of points positioned in three-dimensional space in the form of (x,y,z) coordinates, where z is the depth within the two-dimensional image. Depending on the use case and the capture device points can be associated with other data such as colors and reflectance properties [26a]. Point clouds are captured using specialized devices such as LiDAR(Light Detection and Ranging) cameras that project laser pulses and measure return time in order to provide the third dimension of the points [26b]. There are other technologies for achieving this result but are out of the scope of this work. LiDAR cameras are mostly utilised in cars for achieving self-driving capabilities. Other scenarios are outside of the scope of this work.

The most popular file formats for point clouds are .PLY which is a polygon file format (binary and ascii text variants) and .LAS for LiDAR data. This is where i can mention the intel realsense contribution.

Object detection is a domain of computer vision that attempts to classify objects within an image and find their bounds. In the case of 2D object detection it is about providing a rectangle and in Point Cloud (PC)s it is about providing a cube.

Face detection is a specific case of object detection that is useful in different scenarios. Not to be confused with face recognition, which is about comparing a face scan against the scans from a face recognition database [SBJ17]. In that domain there is more focus about extracting facial features and trying to discern different faces. In this work we only explore object and face detection and not face recognition.

Point clouds produced by LiDAR cameras are around 100k points per frame, a number changing based on the technical specification of the camera. In some cases point clouds are converted into a three-dimensional grid comprised of cubes that each contains a certain number of points. These cubes are also known as *voxels* in the literature and are primarily

used for reducing the computational cost of operations on PCs and grouping neighboring points.

Add some extra images.

2.1.2 Object detection in point clouds

Object detection in PCs imposes extra constraints on the AI architectures developed for learning based tasks. The model result should ideally follow some prescriptive rules in order to be effective at the task at hand as outlined in [Qi+17a]. Those rules are implicitly followed in all methodologies presented in this section.

1. The ordering of the points should not influence the end result.

2. Transformations of the points, such as translation(moving the points around together) and rotation should not change the final prediction.

3. The neighborhood of a point contains significant information that should be actively utilised in the learning process.

The model should be able to handle sparse point clouds, as they commonly occur when doing captures using LiDAR cameras.

Foundational work in object detection for PCs has been done in [Qi+17a]. In this work the authors introduce PointNet, a neural network architecture that operates on PCs and not some other derivative form as we will see in other approaches. They approximate a symmetric function by utilising a multi-layer perceptron (MLP) and a max-pooling function and therefore achieving the first model requirement as outlined earlier. The work in this model and the refined version in the next paragraph is reused in newer architectures.

The authors of PointNet followed up on their work and presented [Qi+17b] where they introduce an enhanced version of PointNet, called PointNet++. In this work they replaced the single max pooling operation of PointNet with a hierarchical mechanism that samples centroids from the point cloud, groups the points in regions defined by the centroids and then extracts local features from the regions. In this way, patterns are captured in multiple scales making features more detailed and robust.

In [ZT18] the authors take a different approach and instead of raw-point clouds their model, VoxelNet, uses voxels as the data representation, allowing them to take advantage of more points in a point cloud efficiently. More specifically they introduce a voxel feature encoding layer(VFE) layer that summarizes point-wise features in a voxel. Stacking VFE

layers achieves a high-dimensional representation, which is then used by a region proposal network(RPN) for object detection.

In [SWL19] the authors propose a 2 stage object detection methodology based on R-CNNs and PointNet++ as a backbone network. The first stage generates a small number of 3D bounding boxes and separates points into foreground and background ones. The second stage then refines those proposals using the information produced in the first stage.

The work in [SR20] brings the graph neural network(GNN) architecture into point clouds. It constructs a graph of the point cloud by adding edges between neighboring points in a fixed radius. For the detection it uses a MLP for classification and a MLP that computes a bounding box for each class.

the work in [Pan+21] introduces PointFormer, a Transformer based backbone for learning features from point clouds. To capture local, global and the interplay between local and global information the authors create a local Transformer(LT) module, a global Transformer (GT) module and a local-global transformer (LGT). The LT module learns features in small regions, the GT module learns context-aware representations at the captured point cloud level while the LGT aggregates local features into the context used by the global transformer.

2.1.3 Person/Face detection in point clouds

Face detection as a specific task is not that popular in the point cloud space. Face recognition has been covered in more existing work as point clouds offer richer information than the conventional 2D images. In contrast, people detection has been a topic of interest in a lot more work, as it is massively more represented in datasets and important for self-driving related tasks. In [JL21] the authors present a single stage 3D person detection network that uses a submanifold sparse convolution as an attempt to evaluate its effectiveness in this domain. In [Bel+25] the authors implement people detection by using a Random Forest Classifier(RFC) based on hand-crafted features such as voxel-based curvature histograms and local surface curvature variations. The point clouds are first pre-processed by removing the ground points and thus keeping points richer in information. As we will see in the datasets section most datasets include the category *pedestrian* which is a less invasive process in contrast to faces.

2.1.4 Datasets for point clouds object detection

For face detection extra processes are required to get access to the datasets. Datasets containing faces are mostly for tasks such as face recognition, pose estimation and other

tasks that require more tuning than conventional object detection. They differ significantly from object detection datasets as they already have cropped and aligned the faces in the data without leaving the rest of the background intact. Most datasets for object detection in point clouds are for scenarios involving self-driving vehicles and are captured by LiDAR devices. They have pedestrians as a category but not faces.

Several large-scale datasets have been proposed for this domain. The work at [Gei+13] created KITTI, one of the earliest benchmarks for LiDAR-based 3D object detection, annotating classes such as car, pedestrian, and cyclist. In [Mao+21] the authors present ONCE, a dataset with 1 million LiDAR scenes and annotations for 5 classes. The Argoverse 2 dataset is introduced in [Wil+23], notable for its broad 26-category taxonomy, that extends beyond common driving objects to include rare classes such as strollers, wheelchairs, and animals.

The JRDB dataset introduced in [Mar+23] contains around an hour of sensor data in an uncontrolled environment, including data from LIDAR channels and an RGBD camera, produced from a mobile robot. The dataset was then annotated with 3D bounded boxes by human labelers for pedestrians.

2.2 Related Work

In [Ull+22] the authors implement a real-time system for human face detection and recognition. This case involves CCTV images in contrast to our application that is based on point clouds. The face detection module is based on existing 2D based methods, more specifically a cascading classifier. In this work we also utilize a cascading classifer as an alternative mechanism for face detection for when point cloud based methods cannot be reasonably applied. In the work at [Yan+18] the authors use face detection in a method for faster transmission of 3D point cloud data. They do so by nose tip detection(enabled by the face detection module) to find a stable anchor to use for transmitting smaller P-frames. The authors in [Ohn+23] use colorless point clouds instead of conventional cameras to preserve privacy while doing pedestrian trajectory tracking. Their choice of hardware and configuration is what allows them to create a private testbed. The authors of [ZZD24] have created an end to end pipeline for protecting pedestrian privacy in 2D videos. They achieve this by swapping the detected faces with dissimilar but relevant faces from an existing face library combining multiple AI models in order to have a consistent and smooth anonymized video as a result. In this work we attempt to work with point clouds as our data where an analogous process is not directly applicable.

Commercial software found at: `https://www.riegl.com` and at `https://www.celantur.com/`.

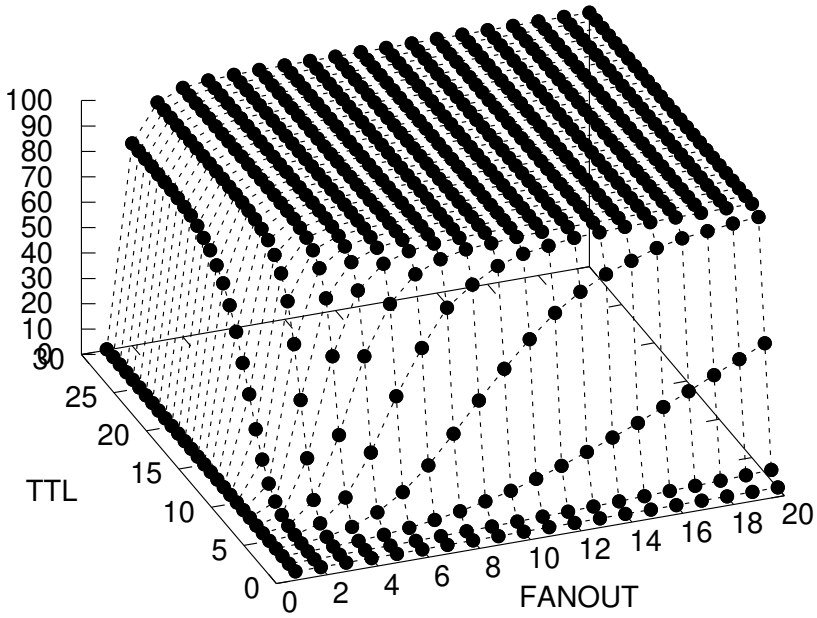


Fig. 2.1: Caption...

3 System Design and Implementation

3.1 Design

- Item 1
- Item 2
- Item 3
- Item 4

3.2 Implementation

Algorithm 1 Test

```
procedure GET(a, b)                    ▷ The g.c.d. of a and b
    r ← a mod b
    while r ≠ 0 do                     ▷ We have the answer if r is 0
        a ← b
        b ← r
        r ← a mod b
    return b                           ▷ The gcd is b
```

4 Evaluation

4.1 Experimental setup

4.2 System evaluation

In this section we present the performance evaluation of the proposed system.

4.2.1 Performance and cost evaluation

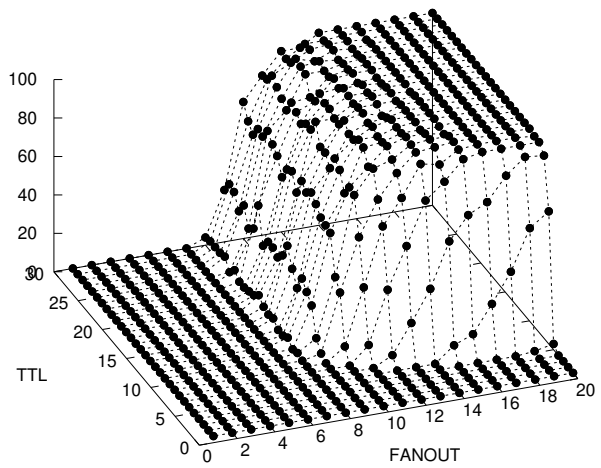
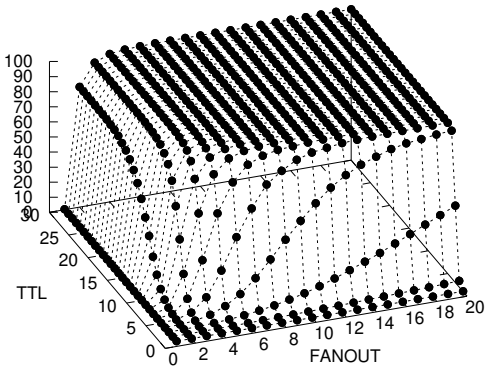


Fig. 4.1: This figure shows the percentage of complete disseminations.

4.2.2 Qualitative evaluation

First column	Second column	Third column
1	2	3
4	5	6
7	8	9

Tab. 4.1: Test table



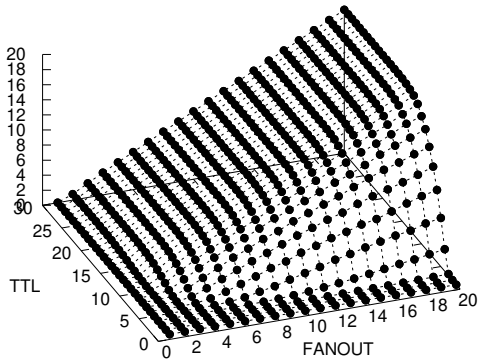


Fig. 4.2: These are the remaining two figures.

4.2.3 Effect of other parameters

Conclusions

5

5.1 Limitations

The 3D Object Detection python ecosystem is not that mature yet. There are great but unmaintained projects unsuitable for software in production environments. More specifically, python versions are outdated, only specific versions of packages are supported. Pretrained and actually useful models are rare in contrast to 2D where for example opencv offers out of the box several object and face detection modules.

Face detection as a specific task is not that popular in Point cloud related settings. Mostly pedestrians in an outdoor setting are actually the target of the latest models. There has been more focus in face recognition tasks as point clouds can provide richer features of a face than the two-dimensional equivalent and can therefore improve the effectiveness of face recognition systems.

Most of the attention is on traffic settings, other use cases remain mostly unexplored e.g. indoors 3D cameras. In [Wan+22] the authors improve upon an existing model to make it more suitable for complex traffic conditions.

5.2 Discussion

5.3 Future Work

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List of Acronyms

API Application Programming Interface

WWW World Wide Web

PC Point Cloud

MLP multi-layer perceptron

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